

EAR INFECTIONS

Key Family Physician: Dr Tang Xin Kai

Specialist Advisor: Dr Goh Xue Ying (Consultant, Department of Otolaryngology – Head & Neck Surgery, NUH)

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MIDDLE EAR INFECTIONS

Introduction

- Otitis media (OM) encompasses a spectrum of diseases, including acute otitis media (AOM), otitis media with effusion (OME) and chronic suppurative otitis media (CSOM).
- Acute otitis media (AOM) is a self-limiting infection that mainly affects children. It is attributable to eustachian-tube dysfunction, most often caused by an upper respiratory infection, and presents with acute onset otalgia and fever.
- CSOM and OME both may occur after an episode of AOM. CSOM is characterized by chronic suppurative middle ear inflammation, usually associated with a persistently perforated tympanic membrane. OME is a chronic inflammatory condition characterized by the presence of an effusion, glue-like fluid behind an intact tympanic membrane without signs and symptoms of acute inflammation.
- This guideline will mainly focus on AOM.

Epidemiology

- AOM is one of the most common diseases in childhood, with approximately 80% of all children experiencing a case of otitis media during their lifetime.
- It occurs at all ages but most commonly between 6 and 12 months of age.

Microbiology

- AOM can be caused by viruses, bacteria or coinfection and it is difficult to distinguish between these.
- The most common pathogens causing AOM include:
 - Bacteria: *Streptococcus pneumoniae*, non-typeable *Haemophilus influenzae*, *Moraxella catarrhalis*
 - Viruses: respiratory syncytial virus (RSV), coronaviruses, influenza viruses, adenoviruses, human metapneumovirus
- In infants under 2 weeks of age, AOM also may be caused by group B *Streptococcus*, enteric gram-negative bacilli, and *Staphylococcus aureus*.

Clinical Presentation

History

1. Otalgia: Ear pain is the most common complaint and the best predictor of AOM. It may manifest as ear rubbing or tugging in a non-verbal child.
2. Decreased hearing
3. Ear drainage: If the tympanic membrane has ruptured, there may be a sudden relief of pain with purulent otorrhoea.
4. Fever: Occurs in one to two thirds of children. High fever is uncommon in adults and may suggest complications.
5. Symptoms of the preceding upper respiratory tract infection
6. Young children, particularly infants, may present with nonspecific symptoms and signs such as fussiness, poor sleep, poor feeding and vomiting.

Physical Examination

1. Key features include:
 - Bulging tympanic membrane
 - Decreased or absent mobility of the tympanic membrane when pneumatic pressure is applied using a pneumatic otoscope (**not done in polyclinic**)
2. Other features which may be present:
 - Partial or complete opacification of the tympanic membrane
 - Erythema of the tympanic membrane
 - Acute tympanic membrane perforation with purulent otorrhoea
 - Bullae in the tympanic membrane

Note: In some patients with suspected AOM, the ear canal may be partially or completely obstructed by cerumen, preventing visualization of the tympanic membrane. In such patients, ear irrigation for cerumen removal should be avoided because of risk of tympanic membrane rupture. If the cerumen cannot be safely removed via curettage or aspiration under direct visualisation, referral to an otorhinolaryngologist (ENT) is appropriate.



Tympanic membrane erythema and bulging.
Adapted from B. Welleschik, Wikimedia
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Investigations

1. Investigations are usually unnecessary for AOM.
2. Imaging studies e.g. contrast-enhanced computed tomography of the temporal bones may be required if complications are suspected.
3. Culture of the middle ear fluid may be required for neonates, infants and children with AOM who appear severely ill, or for immunocompromised individuals.

Diagnosis

Otitis media is diagnosed clinically based on the history and physical findings on otoscopy.

Diagnostic action statements from the American Academy of Paediatrics (AAP) specify that:

- AOM should be diagnosed when there is moderate to severe tympanic membrane bulging or new-onset otorrhoea not caused by acute otitis externa.
- AOM may be diagnosed in children who present with mild tympanic membrane bulging and recent onset of ear pain (less than 48 hours) or intense tympanic membrane erythema.
- AOM should not be diagnosed when pneumatic otoscopy and/or tympanometry do not show middle ear effusion.

Management

Most children and young people get better within 3 days without antibiotics. On the other hand, as AOM is unusual in adults and complications may be significant, antibiotics are the mainstay of treatment of uncomplicated acute otitis media (AOM) in adults.

1. Symptomatic medications

- Pain relief: Regular doses of paracetamol or a non-steroidal anti-inflammatory medication (NSAID).
- Decongestants and antihistamines do not help **with** symptoms.

2. Treatment with antibiotics depends on the age and severity

- Severe AOM – defined as AOM with moderate or severe otalgia, or otalgia for at least 48 hours, or temperature 39°C or higher
- Non-severe AOM – AOM without severe signs or symptoms i.e. mild otalgia for less than 48 hours, and temperature less than 39°C

Age	Clinical Status / Presentation	Treatment
Children 6 months and older	Severe AOM (bilateral or unilateral)	Treat with antibiotics
Children aged 6 to 23 months	Non-severe bilateral AOM	
Children aged 6 to 23 months	Non-severe unilateral AOM	Either • Treat with antibiotics, OR • Watchful waiting, with initiation of antibiotics if the child worsens or fails to improve within 48-72 hours of symptom onset based on joint decision making with the caregiver
Children 24 months or older	Non-severe AOM (either unilateral or bilateral)	
Adults	Any severity of AOM	Treat with antibiotics

- Choice of antibiotic, when a decision to treat with antibiotics has been made:

Age	Choice	Antibiotic
Children and young people	First-line if - No amoxicillin use in the last 30 days - No concurrent purulent conjunctivitis - Not allergic to penicillin	Amoxicillin 40-50 mg/kg/day divided in 2 or 3 doses; maximum of 1.5 g/day
	First-line for penicillin allergy	Clarithromycin 15 mg/kg/day divided in 2 doses; maximum of 1 g/day
	Second-choice if - The patient has received amoxicillin in the last 30 days - Presence of concurrent purulent conjunctivitis - History of recurrent AOM unresponsive to amoxicillin	Amoxicillin/clavulanic acid 40-50 mg/kg/day (amoxicillin component) divided in 2 doses; maximum of 3g/day (amoxicillin component)
Adults	First-line	Amoxicillin/clavulanic acid 625mg TDS
	First-line if allergic to penicillin	Doxycycline 100mg BD
	If allergic to penicillin and has contraindications to doxycycline	Clarithromycin 500mg BD

- Duration of antibiotic treatment:
 - Children
 - 10 days – if age <2 or with tympanic membrane perforation
 - 5-7 days – if ≥2 years with intact tympanic membrane and no history of recurrent AOM
 - Adults
 - 5-7 days – for mild to moderate infections
 - 10 days – for severe infections (significant hearing loss, severe pain, and/or marked tympanic membrane erythema).
- Reassess the patient if symptoms worsen significantly or fail to respond to the initial antibiotic treatment within 48 to 72 hours.
- Consider:
 - Alternative diagnoses, such as otitis media with effusion

- If there are symptoms or signs suggesting a more serious condition
- AOM due to resistant organisms, especially if the patient had recently received beta-lactam antibiotics or has recurrent AOM

If the patient is at high risk of infection with a drug resistant organism and did not respond to initial treatment, consider higher doses:

- Children: amoxicillin 80-90mg/kg/day **divided** in 2 to 3 doses
- Adults: amoxicillin/clavulanic acid with the amoxicillin component being 80mg/kg/day. This may require a high dose formulation e.g. amoxicillin 2000mg/clavulanate 125mg BD, which is not available in polyclinic.

Preventive Care

- All children should be recommended to receive the pneumococcal conjugate vaccine and the annual influenza vaccine according to the immunization schedule.
- Avoid tobacco smoke exposure.
- Prophylactic antibiotics are **not** indicated to reduce the frequency of episodes of AOM in children with recurrent AOM.

When To Refer

- Refer patients to the emergency department if they have AOM associated with:
 - a severe systemic infection
 - acute complications, including mastoiditis, meningitis, intracranial abscess, sinus thrombosis or facial nerve paralysis.
- Refer children and adults with recurrent AOM (3 episodes in 6 months, or 4 episodes in 12 months with 1 episode in the preceding 6 months) to ENT. Clinicians may offer tympanostomy tubes for children with recurrent AOM. Adults with recurrent unilateral AOM may require investigation for eustachian tube or nasopharyngeal pathology.
- Hearing loss that persists for more than 2 weeks following resolution of the infection and effusion is abnormal and should be followed up with an audiogram and ENT consultation.
- Patients with tympanic membrane perforations that persist for 12 weeks or longer should be referred to ENT for further management.

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EXTERNAL EAR INFECTIONS

Introduction

- Otitis externa (OE) refers to inflammation of the external auditory canal, which may also involve the pinna or tympanic membrane. It can be infectious or non-infectious.
- OE can be classified as acute (lasts less than 6 weeks) or chronic (lasts more than 3 months).
- Acute otitis externa (AOE) is most commonly caused by a bacterial infection and presents with rapid onset of ear pain, tenderness, itching, aural fullness and hearing loss.
- Necrotising otitis externa infection (also called malignant otitis externa) involves skin and soft tissue of the external auditory canal and bone tissue of the temporal bone.

Epidemiology

- OE is a common condition. Approximately 10% of people will develop otitis externa during their lifetime, and most cases (95%) are acute.
- While OE can occur at any age, the peak incidence is at around age 7 to 14 years. It is uncommon in patients younger than 2 years old.

Microbiology

- The most common pathogens responsible for AOE are *Pseudomonas aeruginosa* and *Staphylococcus aureus*, often occurring as a polymicrobial infection.
- Fungal otitis externa is rare, accounting for about 9% of ear canal infections. It can occur as the primary infection or can develop along with bacterial external otitis, usually after treatment with antibiotics.

Clinical Presentation

History

Symptoms of AOE

- Ear pain
- Pruritus
- Otorrhoea
- Hearing loss
- Aural fullness

Presence of risk factors

- Water exposure, especially swimming
- Trauma or use of devices in the ear canal (cotton swabs, earplugs, hearing aids)
- Skin conditions e.g. atopic dermatitis, psoriasis, allergic contact dermatitis
- Narrow external ear canals
- Ear canal obstruction (cerumen obstruction, foreign body)
- Prior ear surgery or radiation
- Stress
- Immunocompromised patients

Other treatment considerations

- Any prior treatment of OE with topical antibiotics, as persistent or recurrent symptoms may indicate fungal infection

Physical Examination

Examination of the external ear

- Erythema of or signs of trauma to the pinna and tragus
- Tenderness over the tragus, pinna or both

Otoscopic examination:

- Ear canal oedema and erythema
- Associated debris: typically yellow, brown, white, or grey; spores or curdy debris are suggestive of fungal OE
- Tympanic membrane
 - The tympanic membrane may be erythematous in AOE and only partially visible due to oedema of the external auditory canal.
 - The presence of an air-fluid level along the tympanic membrane is indicative of a middle ear effusion and concomitant otitis media.
 - If the tympanic membrane is perforated, AOE is less likely. Instead, it is likely that the primary focus of disease is the middle ear, with secondary inflammation of the ear canal.

Complications of otitis externa

1. Periauricular cellulitis

- Presents with erythema, oedema, and warmth of the skin around the pinna
- Generally mild pain and without systemic manifestations

2. Malignant external otitis (or necrotizing external otitis)

- A potentially fatal complication of acute bacterial OE
- Most common in older adult diabetic patients or other immunocompromised individuals
- Occurs when the infection spreads from the skin to bone and marrow spaces of the skull base
- Patients present with otorrhoea and severe otalgia that appears out of proportion to examination findings.
- Examination findings: Granulation tissue at the bony cartilaginous junction of the ear canal floor; frank necrosis of ear canal skin may be present.
- Cranial nerve palsies, if present, are a poor prognostic sign

Investigations

- Investigations are usually not required for acute otitis externa.
- Cultures may be done for patients with severe, recurrent or chronic external otitis.
- Patients with suspected malignant otitis externa may require magnetic resonance imaging or a computed tomography scan.

Diagnosis

The diagnosis of AOE is clinical, based upon a characteristic history and physical examination.

Guidelines from the American Academy of Otolaryngology-Head and Neck Surgery defined the following components of the diagnosis of AOE:

1. Having rapid onset (generally within 48 hours) in the past 3 weeks, AND
2. Symptoms of ear canal inflammation, including otalgia, itching, or fullness, with or without hearing loss or jaw pain, AND
3. Signs of ear canal inflammation, which include tenderness of the tragus and/or pinna, or diffuse ear canal oedema and/or erythema, with or without otorrhoea, regional lymphadenitis, tympanic membrane erythema, or cellulitis of the pinna and adjacent skin.

Differential diagnosis

Clinicians should distinguish AOE from other causes of otalgia, otorrhoea, and inflammation of the external ear canal.

1. Contact dermatitis
 - Contact dermatitis can be commonly caused by ototopical medication, cosmetics, or shampoos.
 - Pruritus is the dominant symptom. There are often extensive secondary changes from rubbing and itching.
 - Contact dermatitis should be considered if there has been a lack of response to appropriate OE treatment over 1 week.
2. Chronic suppurative otitis media (CSOM)
 - Symptoms of CSOM include otorrhoea and possibly otalgia, hearing loss, tinnitus, or vertigo.
 - Ear canal symptoms and signs are usually mild compared with acute bacterial OE. Tympanic membrane perforation and purulent middle ear drainage are typically seen on otoscopy. There is usually minimal oedema of the external auditory canal.

3. Carcinoma of the ear canal

- Cancer of the ear canal should be considered if there is an abnormal tissue growth in the ear canal or a lack of response to prolonged OE treatment.
- Mild pain and bloody otorrhoea are common symptoms. A friable ear canal lesion with surrounding purulence is usually seen on otoscopy.

Management

1. Prescribe analgesia according to the severity of pain.
2. Topical antibiotics should be used for initial treatment of diffuse, uncomplicated AOE.
 - Use a topical antibiotic which provides coverage against *Pseudomonas aeruginosa* and *Staphylococcus aureus*, such as ciprofloxacin ear drops or neomycin-polymyxin B-dexamethasone ear drops.
 - Treat for 1 week. Advise the patient to continue treatment for an additional week if not resolved, and to return for review if symptoms persist beyond two weeks.
 - Instruct patients regarding administration of topical drops (see below).
3. If there is known or suspected perforation of the tympanic membrane, a non-ototoxic preparation should be used. Treatment with a topical fluoroquinolone for 7 days is preferred.
4. Oral antibiotics should be prescribed in combination with topical antibiotics in the following situations:
 - If there is severe OE with cellulitis and/or fever
 - In immunocompromised patients, regardless of severity of infection
5. When oral antibiotics are indicated, the antibiotic of choice is an oral fluoroquinolone. If the patient is not a candidate for systemic fluoroquinolones, amoxicillin/clavulanate is an acceptable alternative.
6. If there is no response to the initial therapeutic option within 48-72 hours, the patient should be reassessed for other causes of illness.

Instructions for Patients

- If possible, get someone to put the drops in the ear canal for you.
- Lie down with the affected ear up. **Put enough drops in the ear canal to fill it up entirely.**
- Once the drops are in place, stay in this position for 3 to 5 minutes. Use a timer to help measure the time.
- A gentle to-and-fro movement of the ear will sometimes help in getting the drops to their intended destination. An alternate method is to press with an in/out movement on the tragus.
- You may then get up and resume your normal activities. Wipe off any excess drops.
- Keeping the ear dry is generally a good idea while using ear drops.
- Try not to clean the ear yourself as the ear is very tender and you could possibly damage the ear canal or eardrum.
- If the drops do not easily run into the ear canal you may need to have the ear canal cleaned by your clinician or have a wick placed in the ear canal to help in getting the drops into the ear canal.

(Adapted from Rosenfeld et. al, 2014)

Preventive Care

- Devices that are used in the ear canal (e.g. hearing aids, ear buds) should not be worn until the symptoms have resolved. They should be disinfected prior to reuse.
- Strategies to prevent AOE are aimed at limiting moisture retention in the external auditory canal and maintaining a healthy skin barrier. These include:
 - Removing obstructing cerumen
 - Using acidifying ear drops shortly before swimming, after swimming, at bedtime, or all three
 - Using ear plugs while swimming
 - Drying the ear canal with a hair dryer
 - Avoiding trauma to the external auditory canal
- No randomized trials have compared the efficacy of different strategies to prevent AOE.

When To Refer

- Patient with refractory symptoms should be started on an oral antibiotic for one week and referred to ENT for further management.
- Patients who may have malignant otitis externa, CSOM or carcinoma of the ear canal should be promptly referred to ENT.

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